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BasicSTARRseq: a Bioconductor R-package for analyzing STARR-seq data

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Introduction

Self-transcribing active regulatory region sequencing (STARR-seq) was first described in 2013 by Arnold et al. [Arnold13]. The method allows to identify and quantify enhancer regions in non-coding DNA in large scale. In [Arnold13] whole genome in *Drosohpila* (as in [Arnold14] and [Shlyueva14]) and in BAC-Regions of human cells. In [Vanhille15] targeted as CapSTARR-seq in murin cells, the method was reviewed in [Muerdter15]. The R-package BasicSTARRseq provides the in [Arnold13] described routine for peak calling in STARR-seq data with an alternativ calculation of the p-Value. It is easy to use.

Methods

In [Arnold13] the method to call peaks i. e. find enhancer regions in STARR-seq data is described but the code was not published.

The peak calling in the package BasicSTARRseq works the following way: All genomic positions having a STARR-seq coverage over a specific quantile (for example 0.9) are considered to be the center of a peak with a specific width (choosen for example in dependence of the average fragment size of the library). If then two ore more potential peaks overlap, the lower one is discarded. To estimate the peaks' reliability, a binomial p-value is calculated. If it is higher than a before choosen specific p-Value the peak is discarded as random variation. There is more than one possibility to choose a binomial model for calculating the p-Value. One model is the following:

- n_1 number of trials = total number of STARR-seq sequences,
- k number of successes = STARR-seq coverage,
- $\hat{p}_1$ estimated sucess probability in each trial = input coverage divided by total number of input sequences.

Another model is:

- n_2 number of trials = STARR-seq coverage plus input coverage,
- k number of successes = STARR-seq coverage,
- $\hat{p}_2$ estimated success probability in each trial = total number of STARR-seq sequences divided by the sum of the total number of STARR-seq sequences and the total number of input sequences.

While the first model emphasizes the position of the peaks the second attaches more importance to the difference between STARR-seq signal and input signal. Due to the fact, that the coverage usally is a lot smaller than the total number of sequences, the p-values of both models are highly correlated in practice, but not equal.

The enrichment of STARR-seq over input coverage is then calculated as follows:

$$\frac{\frac{\text{STARR-seq coverage of peak}}{\text{total number of STARR-seq sequences}}}{\frac{\text{input coverage of peak}}{\text{total number of input sequences}}}$$

In [Arnold13] it seems that the numinator and denuminator are corrected conservatively to the bounds of the 0.95 binomial confidence interval corresponding to model 1 in this formula and the p-value is calculated corresponding to model 2. To let the decision to the user, the package BasicSTARRseq provides both models to calculate the p-value.

By enrichment value or p-value the user can easily filter the calculated peaks according to reliability or height of peak i. e. activity of enhancer.

The user can choose if the data should be deduplicated before analysis to avoid PCR bias (as in [Arnold13, Vanhille15]) but perhaps also lose some by the construction of STARR-seq duplicated reads. To adopt the method to targeted sequencing, the minimal quantile from the first step of the procedure could be down-regulated in respect to target size, to not take into account every sequenced base pair as peak.

BacisSTARRseq includes test datasets extracted from the published data of [Arnold13].

Results

Parts of the in [Arnold13] published row data was processed with BasicSTARRseq to show the funcionality of the package. Published was a list of peaks of the S2 cells annotated with *fragments* of sample, p-*value* and *foldchange* sample over input.

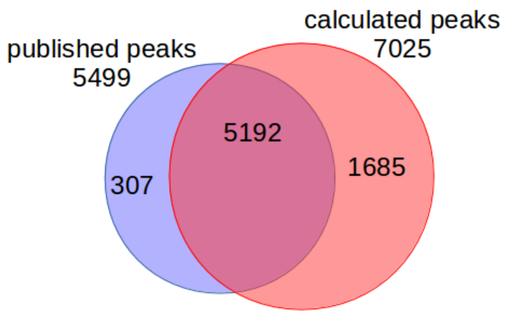

Figure 1: Comparison published peaks and with BasicSTARRseq called peaks.

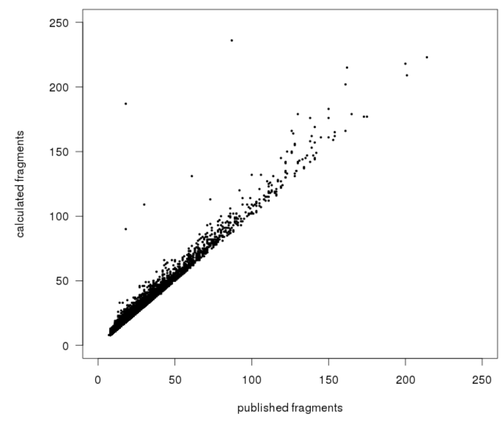

Figure 2: Comparison *fragments* in published peaks and with BasicSTARRsesq called peaks.

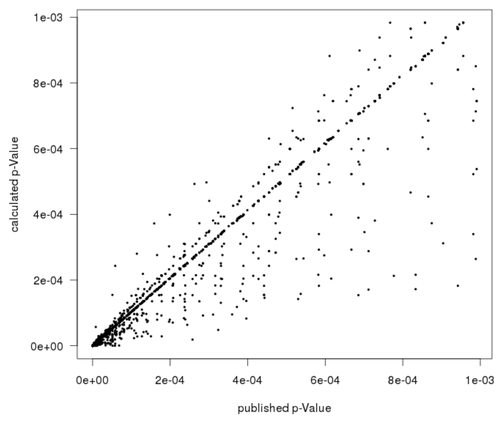

Figure 3: Comparison *p-value* in published peaks and with BasicSTARRsesq called peaks.

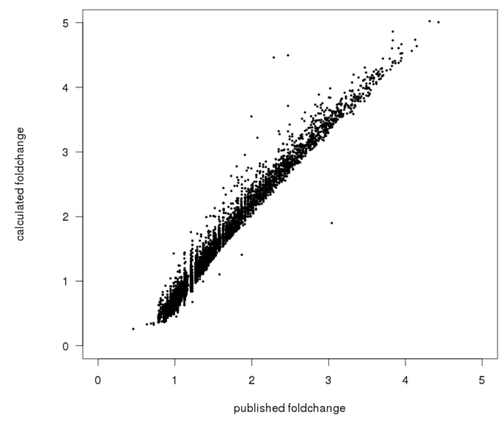

Figure 4: Comparison *foldchange* in published peaks and with BasicSTARRsesq called peaks.

The published peak data could not be reproduced 100% but for the most part. The differences can be due to a slightly different alignment (reference genome dm3) or for example different implementations of the binomial distribution. There are 94% of the published peaks found back with BasicSTARRseq with at least 99% overlap. The correlation than between p-value in these peaks is 0.97, between the foldchange (in logscale) 0.99 and between the fragments 0.98.

Outlook

Quality controls and visualization routines for comparing different replicates, and for comparison of experiment data and target regions are planned for the package. As well plausibility checks or further explorative analysis through comparison functions of range tracks of other analysis (like peak lists of CHiP-seq data, but also other data chosen by the user).

References

[Arnold13] Arnold et al. Genome-Wide Quantitative Enhancer Activity Maps Identified by STARR-seq. *Science.* 2013, 339(6123):1074-1077 **[Vanhille15]** Vanhille et al. High-throughput and quantitative assessment of enhancer activity in mammals by CapStarr-seq. *Nat Commun.* 2015 Apr 15;6:6905. **[Muerdter15]** Muerdter et al. Starr-seq – Principles and Applications. *Genomics* 2015. **[Arnold14]** Arnold et al. Quantitative genome-wide enhancer activity maps for five Drosophila species show functional enhancer conservation and turnover during cis-regulatory evolution. *Nat Genet.* 2014 Jul;46(7):685-92. **[Shlyueva14]** Shlyueva et al. Hormone-responsive enhancer-activity maps reveal predictive motifs, indirect repression, and targeting of closed chromatin. *Mol Cell.* 2014 Apr 10;54(1):180-92.

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